A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

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Motivation

- Climate change-driven large, destructive wildfires becoming more frequent, especially in California [1,2]
- Prescribed burns are known to be effective in reducing wildfire intensity[3], yet much of the existing body of research was performed in the previous century and did not account for climate change's impact on prescribed burn efficacy
- **Statistical analysis** can be used to identify effective or ineffective prevention strategies, such as prescribed burns, but the data needed to study this question is **fragmented** and **complex**

Research Questions

1. Are prescribed burns effective at reducing the size of wildfires?
2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables
3. How do these relationships change at different spatio-temporal scales?

Data

The data set used in this research was created by combining NOAA climate data and CALFire fire history data. This data consists of 2041 wildfires spanning January 2013 to December 2022, where each row is a wildfire and columns include:

- Wildfire acres burned, precipitation, temperature, wind speed, drought index
- For each 3-36 month time frame, prescribed fire acres burned in 5 - 100 km distance radius

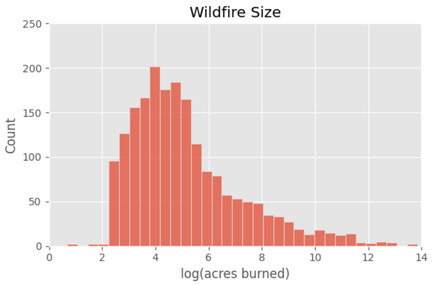


Figure 1: Distribution of wildfire size



Figure 2: Location and size of fires at scale

Table 1: Variables in the dataset

Variable	Source	Description	Range	Units	Symbol
Acres Burned	CALfire[4]	Wildfire acres burned	0+	Log(Acres)	log_acres
Treatment Acres	CALfire[5]	Prescribed fire acres burned	0+	Acres	rx_acres
Precipitation	NOAA[6]	Monthly Precipitation averaged across the county	0+	Inches	pcp
Temperature	NOAA[6]	Average, minimum, and maximum temperature averaged across the county	0+	Fahrenheit	temp
Palmer Index	NOAA[6]	Drought severity indices	[-10, 10]	N/A	zndx, pdsi, phdi, pmdi
Windspeed	OpenMeteo [7]	Maximum windspeed on fire start date	0+	Km/h	max_wind

Methods

Inferential analysis:

- Difference of median test to examine if wildfires are smaller when there is a prescribed fire nearby

Time series analysis:

- Correlation to identify which predictors are associated with acres burned
- Cross-correlation to identify at what time lags predictors are associated with acres burned

Regression Analysis:

- Predict acres burned per wildfire incident from prescribed fires and climate variables

Inferential Analysis

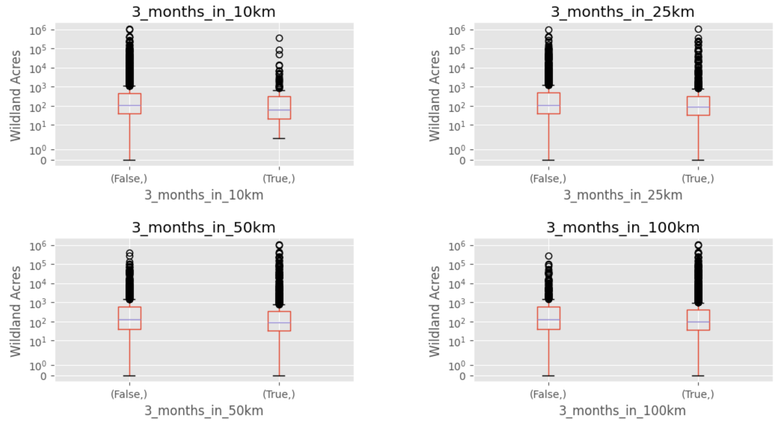


Figure 3: Box plot of wildfire size between groups with and without prescribed burns within the specified distance

Table 2: Mann-Whitney U-test results, including rank biserial correlation (RBC) and common language effect size (CLES)

Month	Spatial Radius	U-value	Tail	P-val	CLES
3	10km	116580.5	Less	**0.002**	0.57
	25km	398966.5	Less	**0.015**	0.531
	50km	47.0224.5	Less	**0.0005**	0.542
	100km	324365	Less	**0.054**	0.527

Mann-Whitney U-test uses ranking to determine if wildfires with prescribed fires near by are smaller on average than wildfires without prescribed fires. The test results in Table 2 yield significant differences, showing prescribed burn treatment is associated with fewer acres burned at 10 km, 25 km, 50 km, and 100km distance when completed 3 months before a fire. The difference can be visually seen in Figure 3, where the median (blue line) is slightly lower for cases with prescribed burns.

Time Series Analysis

Time series analysis relies on the correlation coefficient, R, where $R = 1$ is a perfect positive association and $R = -1$ is a perfect negative association. Results from the correlation study in Figure 4 indicate the amount of wildfire acres burned per month is most strongly associated with climate variables, specifically temperature and precipitation, while prescribed fires are very weakly associated. In the cross-correlation study in Figure 5, the correlation is the strongest for lower lags, indicated climate conditions in the month of the fire are most impactful.

Figure 4: Correlation between wildfire acres burned and predictors

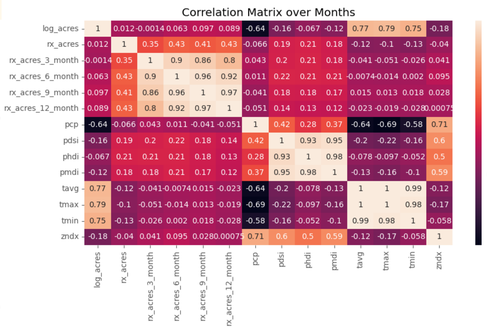
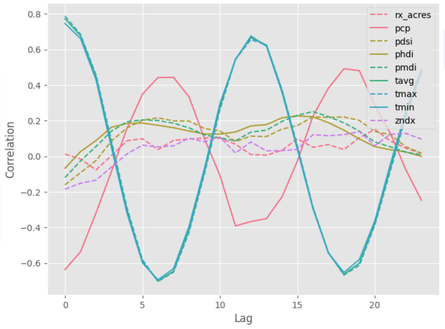


Figure 5: Cross-correlation between wildfire acres burned and predictors



Regression Analysis

Linear regression predicts the size of a wildfire using the prescribed fire acres burned within 3 months and 50 km, previous month precipitation, and maximum windspeed the day of ignition.

Model: $\hat{Y} = \hat{\alpha} + \hat{\beta}\boldsymbol{X}$

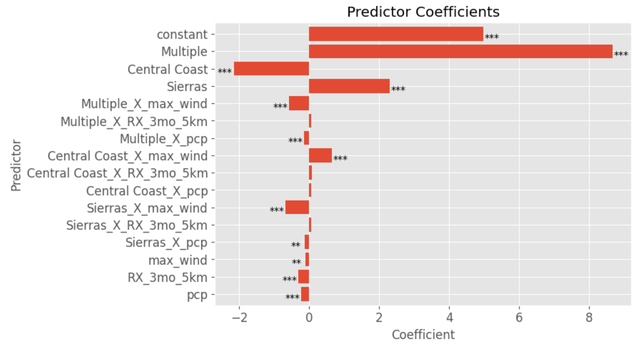


Figure 6: Linear Regression coefficients, with significant predictors notated with "*"

- $R^2 = 0.885$; $F = 747.8, \mathrm{p} < 0.001$
- $Train\ MSE = 3.476$; $Test\ MSE = 4.535$
- Prescribed burns are related to decreases in wildfire size at the 3 month and 50 km spatial-temporal range, seen with the negative coefficient in Figure 6 and negative slope in Figure 8
- Decreased precipitation is related to increases in wildfire size, also visible with the negative coefficient and slope in Figures 6 and 7 respectively
- Region has an effect on wildfire size, with wildfires spanning multiple regions having more acres burned and Central Coast wildfires having fewer acres burned

Figure 7: Partial regression for precipitation

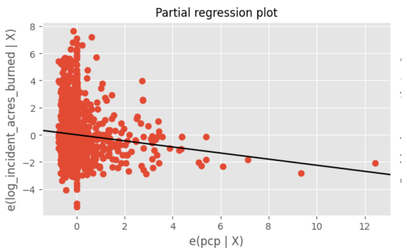
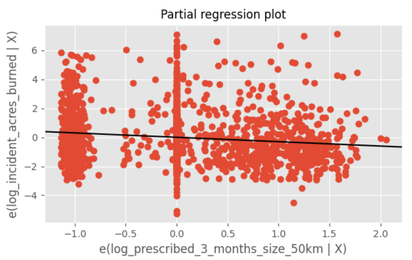


Figure 8: Partial regression for prescribed burns



Conclusion

- Increased prescribed burns are associated with decreased wildfire size for certain cases
- When controlling for climate variables, such as precipitation, prescribed burns are still a significant predictor of wildfire size
- Prescribed burns have the greatest impact when performed within 12 months and 50 km of wildfires when controlling for drought

References

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[2] CALFire. Top 20 Largest California Wildfires. Accessed on 2023-01-05. url: https://www.fire.ca.gov/our-impact/statistics.
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[4] CALFire. Incident Data. 2023. url: https://www.fire.ca.gov/incidents/
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